

Han Yan

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Education

Australian National University (ANU) QS #34 — Master of Computing · ML Specialization

Key courses: Computer Vision, Computer Graphics, Game Development, (Statistical) Machine Learning, Advanced Algorithms, SQL Databases.

Donghua University (211) — Bachelor of Functional Materials

Internships & Work Experience

Hichains | ML / AI Researcher

2025.04 – Present

- **Smart OCR Hub — Lenovo Customs Declaration System (Python / Vision-LLM)**
 - **Dual-path OCR architecture:** primary path runs qwen3-vl-plus full-page OCR + LLM field merging; DocLayout-YOLO + PaddleOCR-VL-1.5 serves as local fallback so the service stays up when the API is unreachable.
 - **Multi-GPU cluster scheduling:** vLLM instances + Redis state bus + SSE for live GPU utilization and alert push, supporting batch-concurrent multi-document processing.
 - **End-to-end MLOps:** MLflow experiment tracking + human review loop + continuous fine-tuning, giving every model iteration an audit trail.
 - **Stack:** FastAPI · SvelteKit · PostgreSQL · Redis · vLLM · MLflow.
- **AURA — Enterprise Multi-Agent Collaboration Platform (Python / SvelteFlow)**
 - **Blueprint orchestration engine:** visual DAG editor built on SvelteFlow turns SOPs into reusable agent workflows, with parallel dispatch, rate-limiting, version snapshots, and replay.
 - **Multi-scenario adaptation:** a unified surface for meetings (live transcription + auto-dispatched minutes), projects (blueprint execution + RAG archival), and management (permission grouping + full audit trail).
 - **Real-time collaboration core:** Socket.IO + Redis pub/sub + Yjs collaborative editing, supporting multi-user multi-Agent parallel tasks on a single screen.
 - **Stack:** FastAPI · SQLAlchemy 2.0 · PostgreSQL · Alembic · SvelteKit 5 · MLflow.

ANU Techlauncher Capstone | Vision Engineer + PM

2024.02 – 2024.11

- **Human Fall Pose Detection & Audio Alert System (Python) — Joint project with a Western Australia company**
 - **Vision algorithm development:** customized YOLOv7 model for real-time human keypoint detection and pose analysis; structural optimization for accurate fall recognition.
 - **Project Management (PM):** led project planning and module assignment, using Jira for team task tracking and progress visibility.

Highlight: Presented at the ANU Techlauncher showcase.

Details: [CAM FALL Project](#)